WisBlock BambuHelper assembly guide for landscape enclosure version

Required parts:

- RAK3312 WisBlock Core module ESP32-S3
- RAK14014 WisBlock TFT touch screen 240x320 pixel
- RAK19007 Wisblock Base Board
- RAK18001 WisBock Buzzer
- Switch for battery disconnect (optional)
- 8 pcs self-tapping screws Ø 2.5mm length 5mm

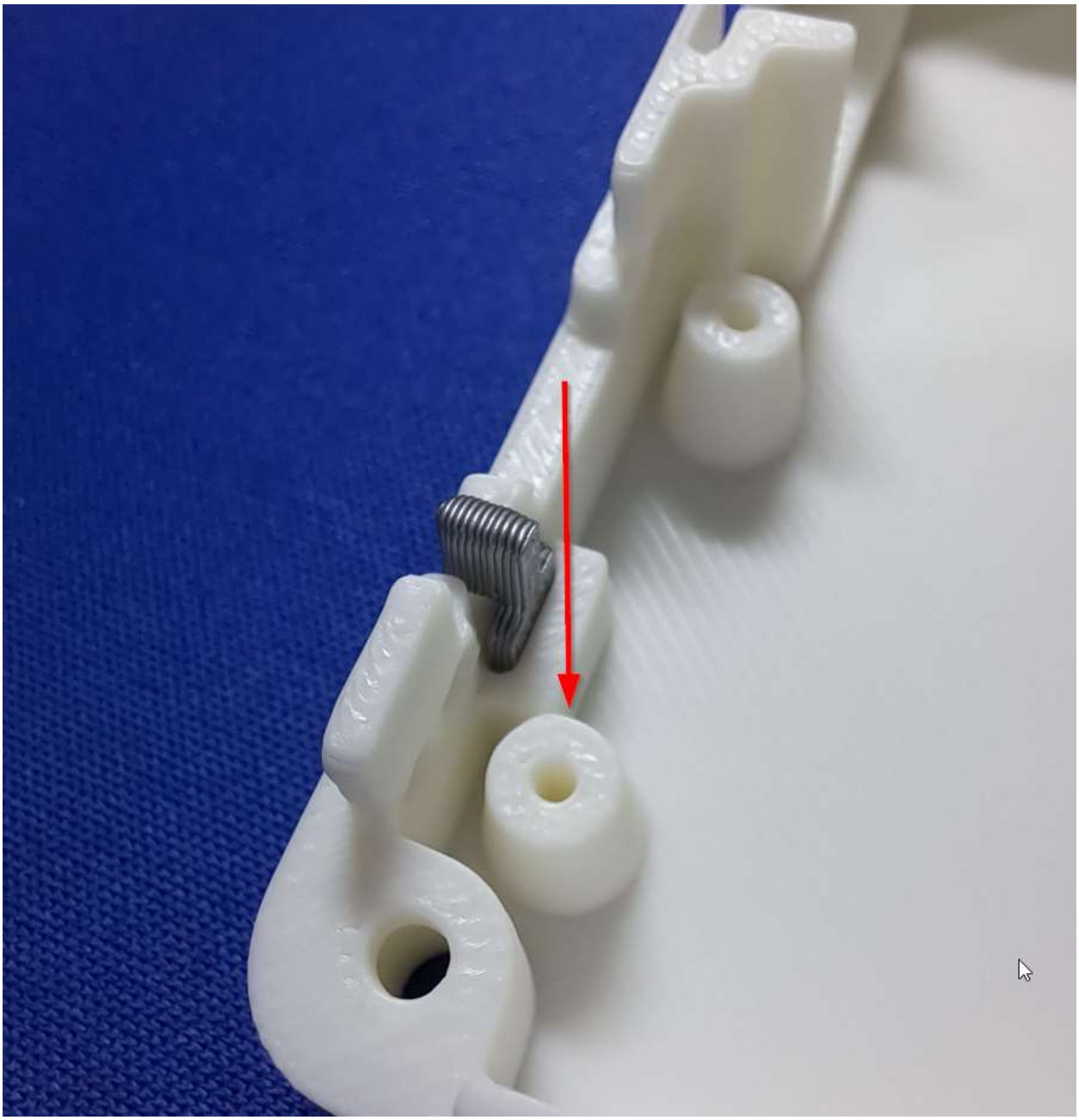
Step 1 – print enclosure

Print the enclosure using the BambuStudio project file or using the single part print files.

If using the single part print files, make sure to add support when printing the lid.

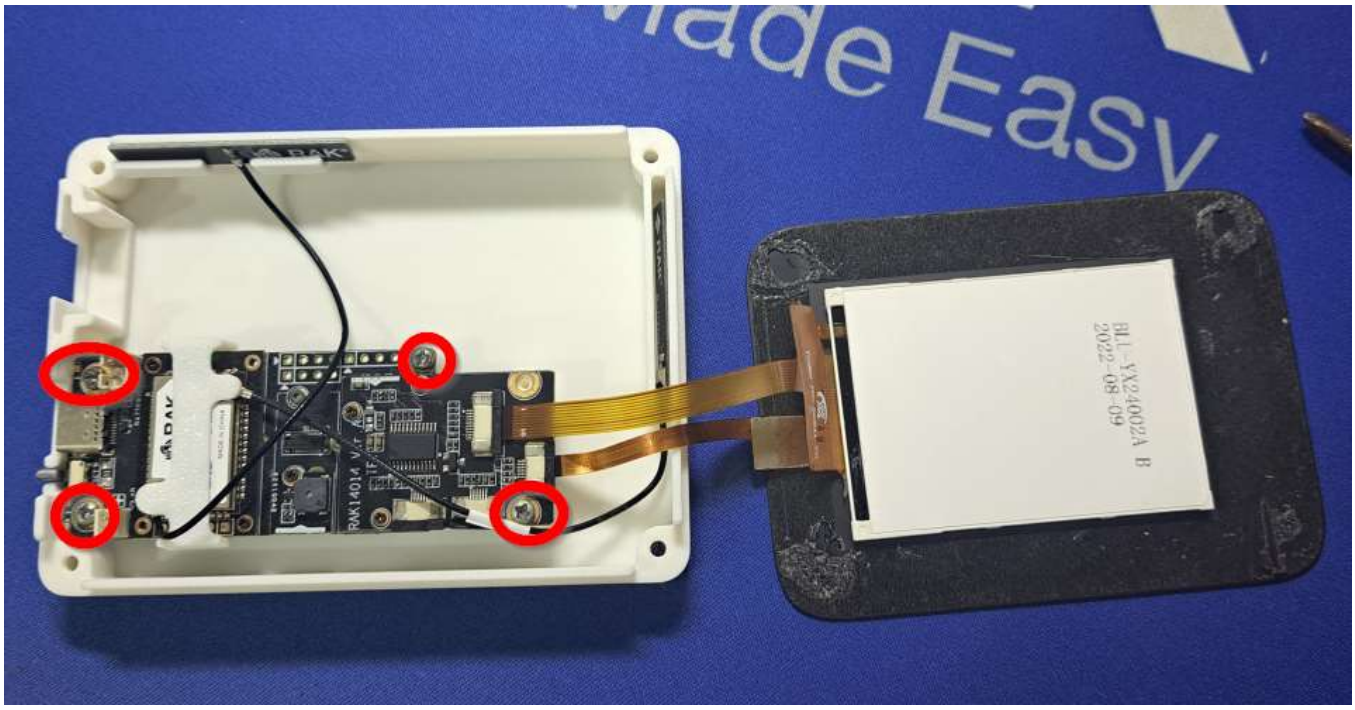
Step 2 – assemble reset button

The reset button is assembled on the left side of the bottom part. It might be difficult to push it in if the electronics is already mounted, so this should be done first.



Step 3 – assembly the WisBlock Modules

Assuming that the TFT interface module, the MCU Core module and the buzzer is already mounted on the WisBlock Base Board, screw the Base Board with 4 self tapping screws

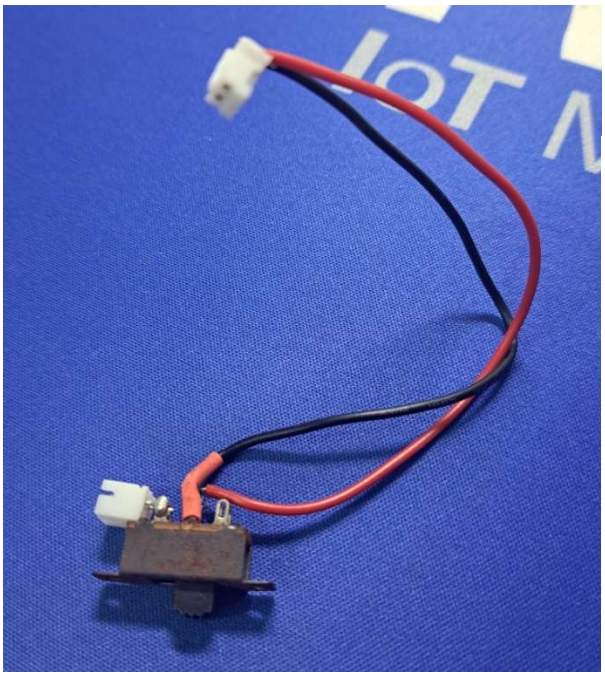


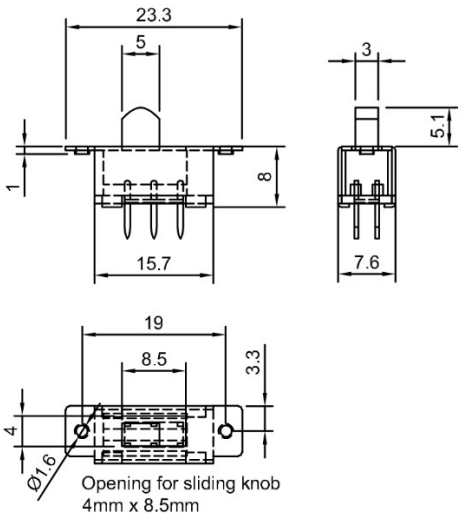
Step 4 – optional, prepare a battery switch

Optional the device can be switched off from battery power using a simple slider switch.

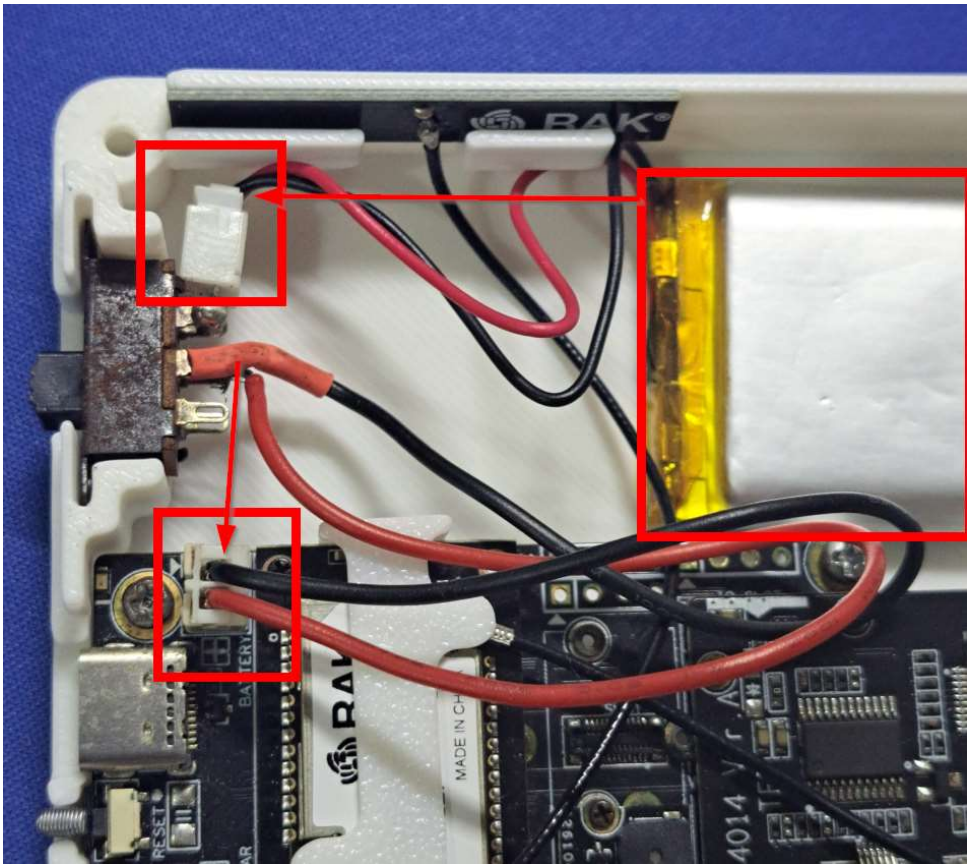
On one side of the switch a connector is soldered to which the battery will be plugged in.

On the other side a battery extension cable is soldered that goes to battery connector on the WisBlock Base Board.



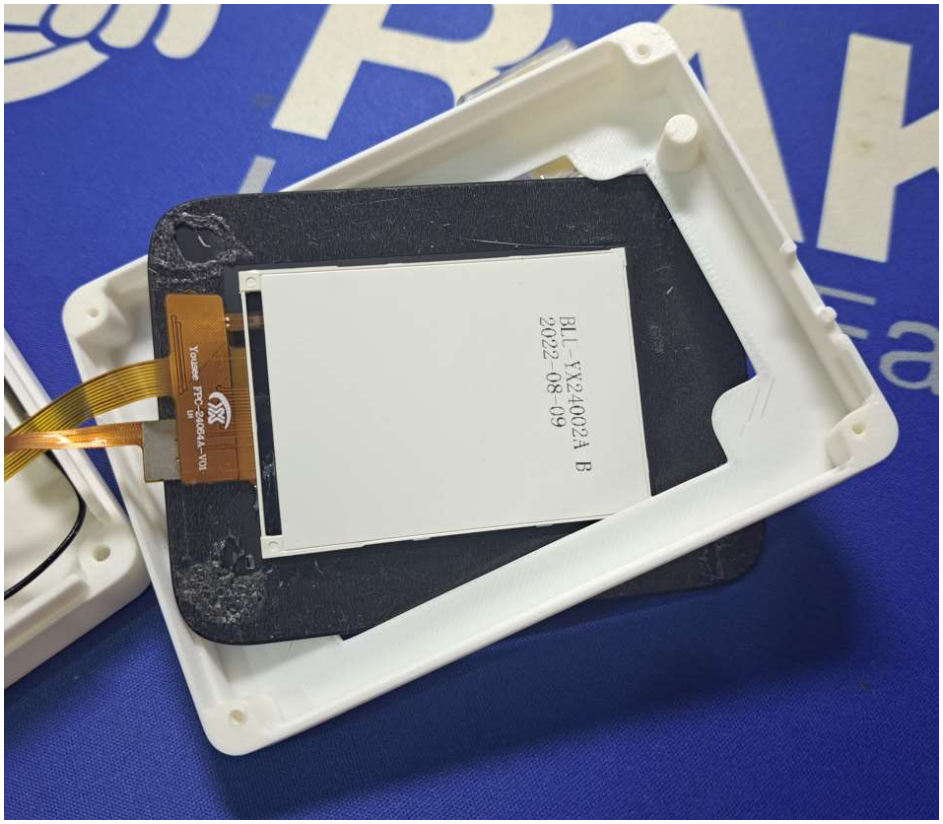


The switch is mounted on the left side of the bottom part



Step 5 – mount the display in the lid

To mount the display, it has to be slid through the center opening of the lid



Then the self-adhesive tape of the display can be used to glue the display to the lid.

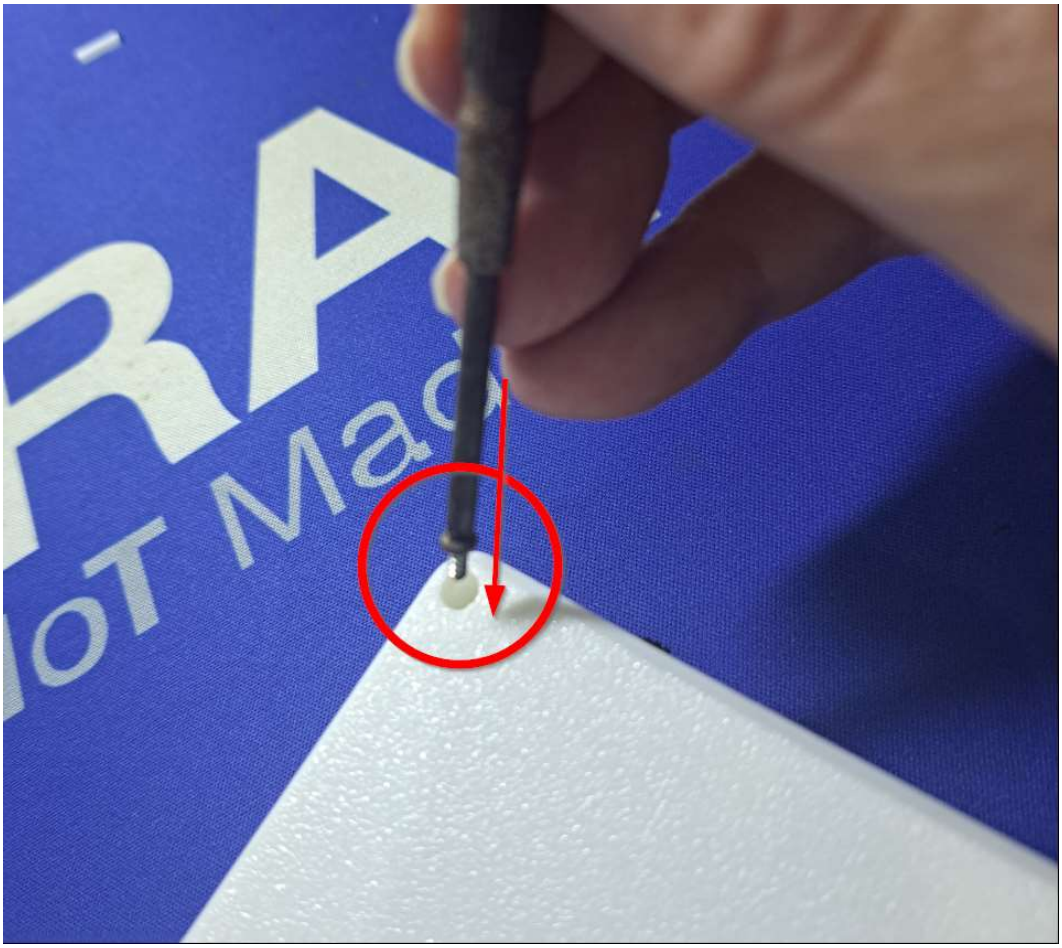
Check that it sits centered in the contour on the top of the lid

Mount the lid on the bottom part. It is best to align the lid first on the right side of the bottom part, then rotate it towards the bottom part. This helps to avoid damage on the sensitive FPC cables between the WisBlock module and the display.

Step 6 – final assembly

Check that the bottom part and lid are sitting correct together

Turn around the device and fix the lid to the bottom using 4 self tapping screws, one in each corner



Step 7 – power up the device and start the setup

To power on the device, plug in a USB cable and move the slider switch to the on position.

Then start to setup the device as explained on KERALOTS Github repo:

https://github.com/Keralots/BambuHelper#setup